

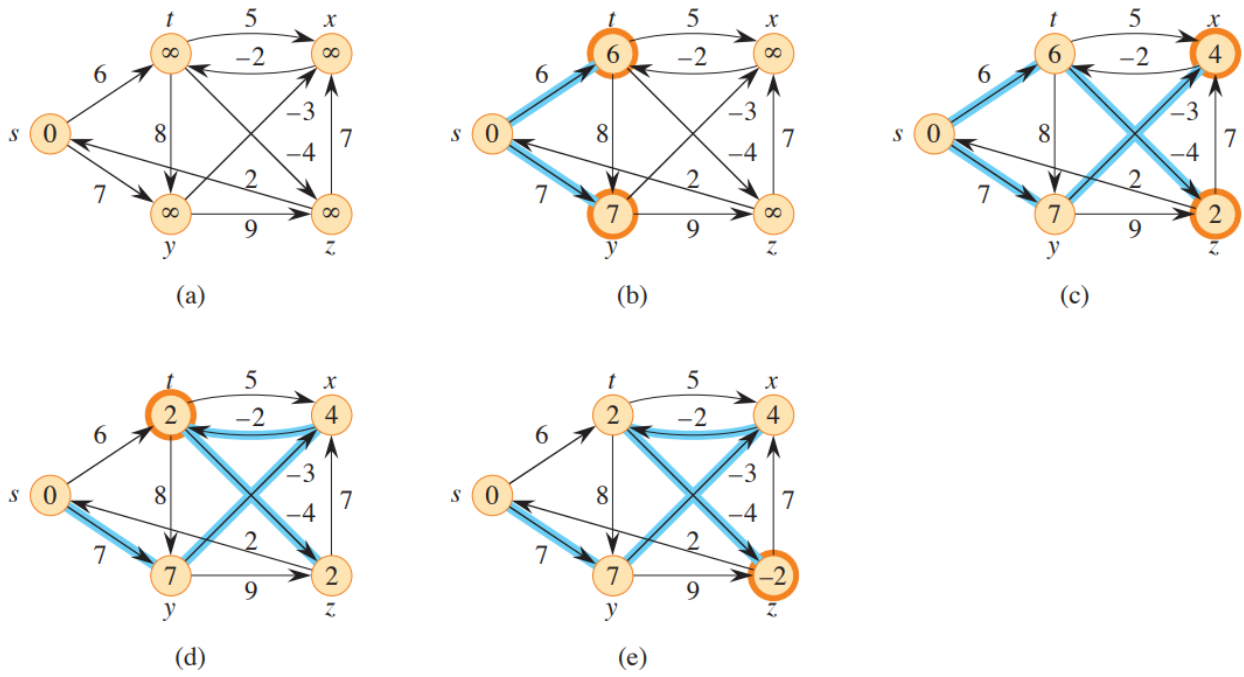


Figure 22.4 The execution of the Bellman-Ford algorithm. The source is vertex s . The d values appear within the vertices, and blue edges indicate predecessor values: if edge (u, v) is blue, then $v.\pi = u$. In this particular example, each pass relaxes the edges in the order (t, x) , (t, y) , (t, z) , (x, t) , (y, x) , (y, z) , (z, x) , (z, s) , (s, t) , (s, y) . (a) The situation just before the first pass over the edges. (b)–(e) The situation after each successive pass over the edges. Vertices whose shortest-path estimates and predecessors have changed due to a pass are highlighted in orange. The d and π values in part (e) are the final values. The Bellman-Ford algorithm returns TRUE in this example.